HALO Design Review

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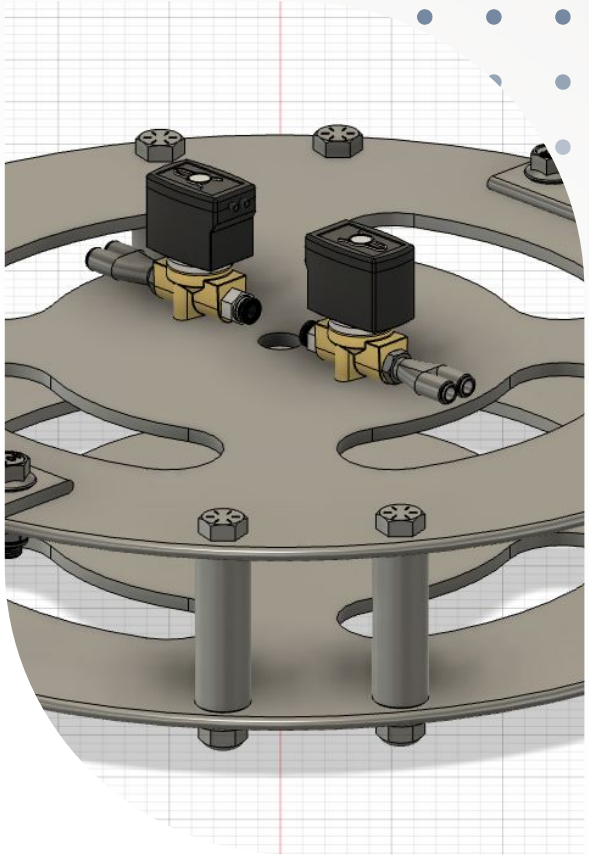
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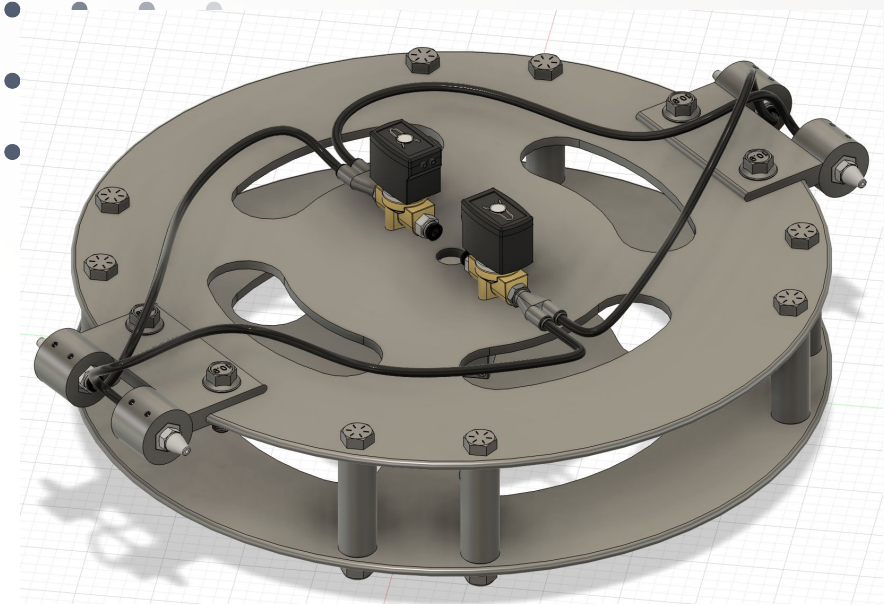
01

Reaction Control System

Mission Success

- The purpose of the cold gas thruster system is to create a efficient roll control system that was well suited for our rockets goal of having precise control to ensure stable flight and landing.

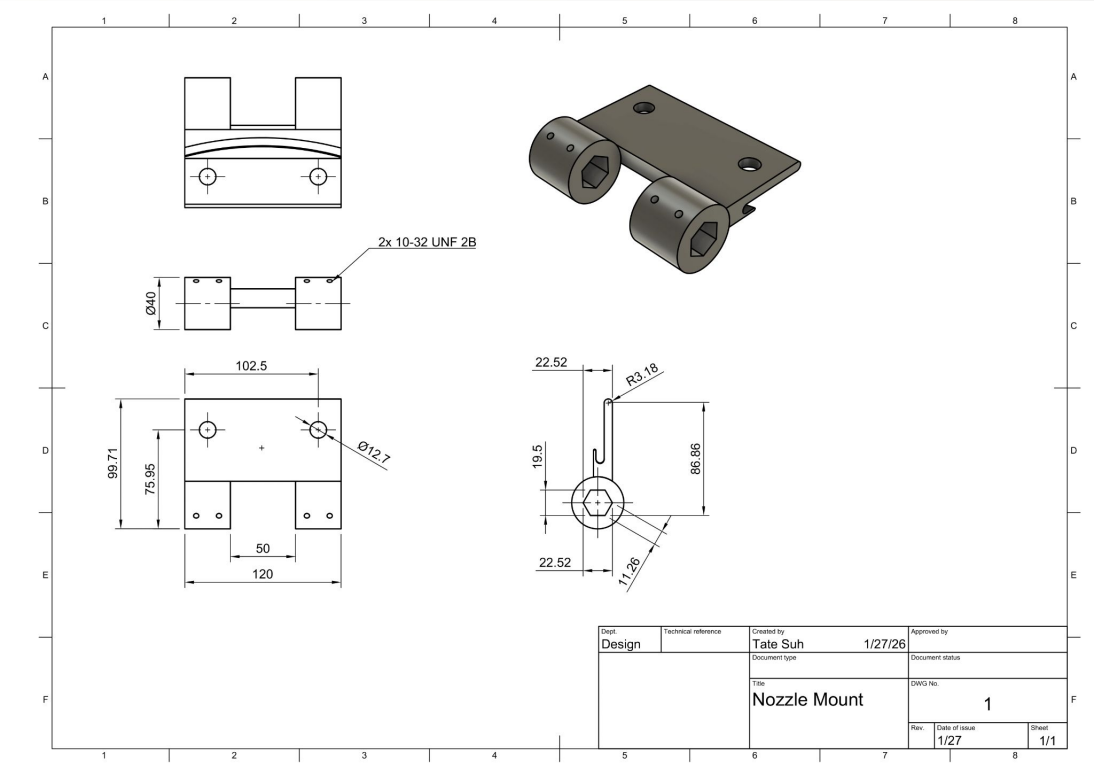
- We decided on a cold gas thruster system for roll control instead of a reaction wheel because of its small footprint and ability to scale with rocket weight easily.



What’s new?

- Mounting methods
- Spacers
- Tubing

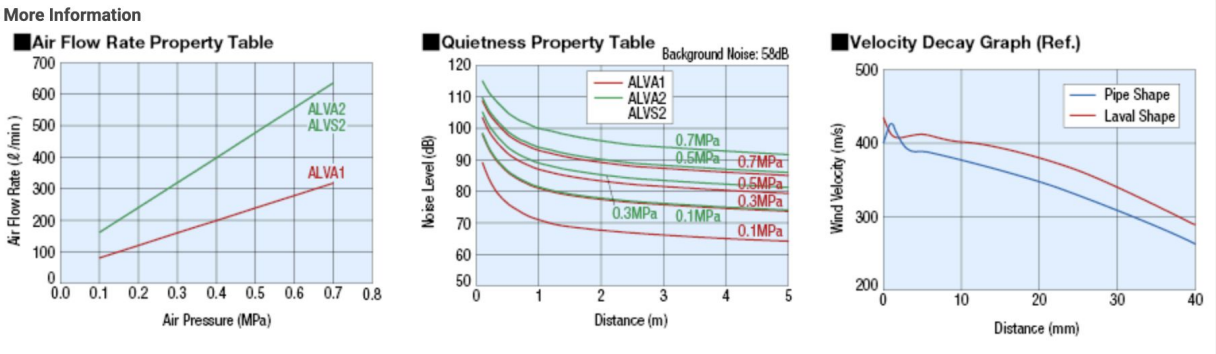
Final Nozzle Mount Design



Material: PETG

Weight: ~150g

Calculations



Using the standard thrust equation $F = \dot{m}V_e + (p_e - p_0)A_e$ and values from the manufacturer, the thrust from one nozzle was calculated to be 5.75N. From this, the estimated torque and maximum angular acceleration are 3.23Nm and 28.17deg/s^2 respectively.

Test Plan Overview



Phase One: Safety & Setup

Assemble the system and bolt down the nozzle housings. Pressurize system and with valves closed and check for leaks and push to connect fittings.



Phase Two: Static Thrust Ver.

Design and build a pivot arm test rig. Test short bursts to record max thrust. Fire long bursts to ensure negligible thermal losses and thrust drops.



Phase Three: PWM Characterization

Identify the maximum switching frequency and linearity of the solenoid valves. Test frequency limits at 50% duty cycle. Use optimal frequency and sweep duty cycle to find dead zones.

BOM

Regulator	KPR1FRB417A20000	$443.78	1	1	Steps down pressure from propulsion systems 1000psi to 100psi	Swagelok
1/4" NPT to 1/4" NPT Female Adapter	N/A	$4.99	2	2	Thread adapter for the nozzles	Amazon
Pressure Relief Valve	48435K72	$6.28	1	1	Relieves pressure when regulator fails	McMaster
Thread Adapter (1/4" mNPT to 3/8" mNPT)	CA-3325CB-4	$9.49	1	4	Thread adapter for after the regulator	Amazon
Arming Valve	DERNORD-476	$11.99	1	1	Turns on and off pressure for RCS	Amazon
Tubing Adapter (1/4" NPT to 1/4" OD)	PC-1/4-N2	$11.99	1	10	Tubing adapter for fittings	Amazon
Tubing Adapter (3/8" NPT to 3/8" OD)	PC-3/8-N3	$12.99	1	5	Tubing adapter for fittings	Amazon
Solenoid Valves	2BCK-1/4-D	$42.73	2	1	Solenoid valves that opens and closes gas to nozzles, used to control RCS	Wic Valve
Wye Adapter	5779K627	$9.78	2	1	Splits gas output from solenoid valves to two nozzles	McMaster

BOM

Tubing (1/4" OD)	N/A	$16.19	1	1	Tubing for gas	Tailonz
Tubing (3/8" OD)	N/A	$20.69	1	1	Tubing for gas	Tailonz
Manifold	1023N15	$24.56	1	1	Splits gas output to both solenoids	McMaster
Nozzle	ALVS2	$22.37	4	1	Nozzles to spit gas out for RCS thrust	Misumi
Thread Adapter (1/2" fNPT to 1/4" mNPT)	421423	$9.99	1	1	Adapts threading for regulator	Fastenal
JIS High-Strength Steel Flanged Hex Head Screw	95735A117	$10.67	1	10	Bolts for plates	McMaster
Medium-Strength Steel Flange Nut-Class 8	90374A117	$8.85	1	25	Nuts for plate bolts	McMaster
18-8 Stainless Steel Cup-Point Set Screw	92311A427	$8.71	1	100	Hold nozzles in place	McMaster
JobSmart 1/4 in. Male NPT to 3/8 in. Male NPT Adapter	1333105	$5.99	1	1	Regulator to arming valve/manifold adaptor	Tractor Supply

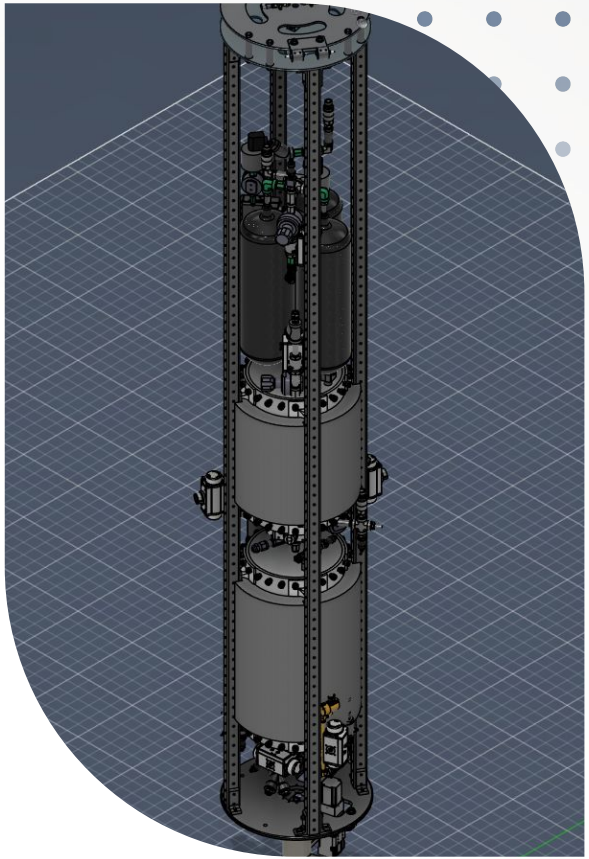
Total: $806.65

Manufacturing and Procurement

- The nozzle mounting can be printed in house
- Plates will be ordered and could potentially be one of our long-lead items.
- We are currently on track to have RCS fully built before the quarter is over and hopefully partially tested as well.
- No current independencies but I do want to know who will be working on RCS with me.

Development Roadmap

- **Phase One: Build (Weeks 1-2)**
  - Assemble the system once all parts have arrived.
- **Phase Two: Component Testing & Verification (Weeks 3-4)**
  - Validate performance models and characterize solenoid behavior
- **Phase Three: System Integration & Validation (Weeks 4-5)**
  - Integrate the propulsion system onto the vehicle frame and verify leak integrity.



02

Structure

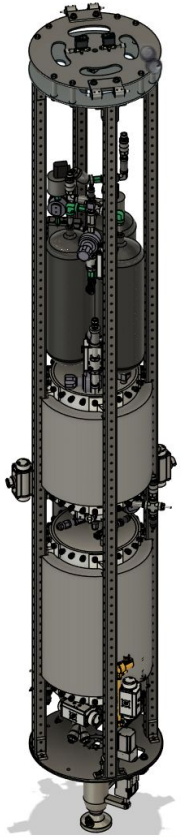
Executive Summary

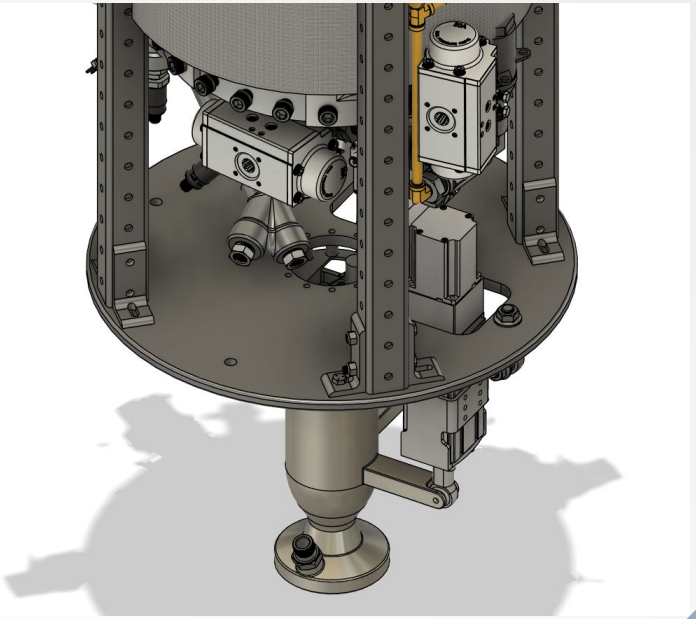
- The Structures subteam enables mission success by providing a lightweight, rigid airframe that maintains subsystem alignment and withstands launch, flight, and recovery loads, allowing all systems to operate reliably throughout the mission.
- Since the previous design review, the structure transitioned to four vertical aluminum rails and a redesigned baseplate as the primary load transfer interface.
- All original structural requirements, including thrust load capacity, factor of safety, and manufacturability have been satisfied by the final design. Verification is supported by analytical calculations and FEA results.

Specifications & Design

The Structure consist of Four aluminum rails where different subsystems can be mounted to either by direct connection or the use of plates.

- Base-Plate ~5.5 Kg
- 4 Aluminum rails ~ 10.8

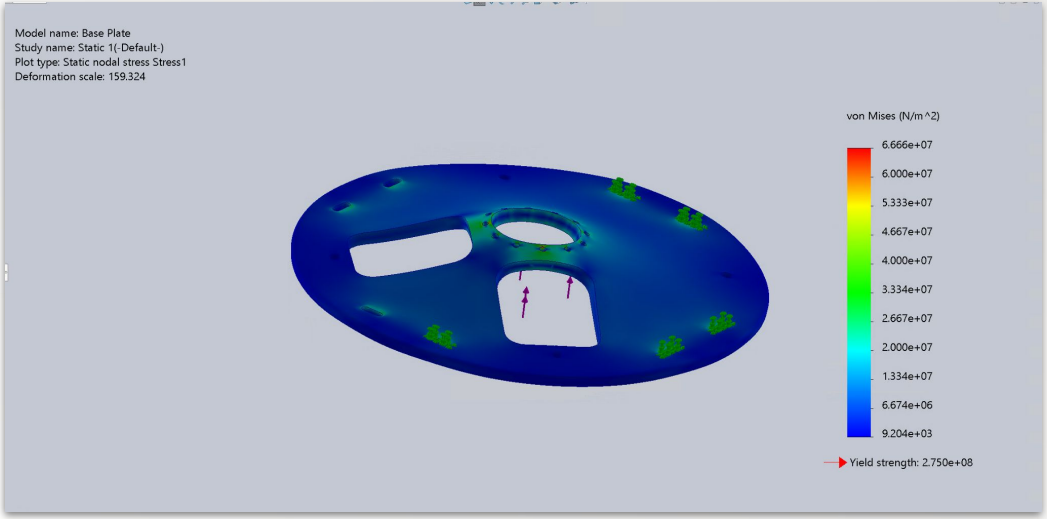




Verification & Validation

Baseplate:

- Material: 6061-T6 Aluminum
- Fixed constraints at rail attachment points
- Loads applied at engine mounting interface



The analysis shows a maximum von Mises stress of approximately 67 MPa near geometric discontinuities. This remains well below the material yield strength, resulting in a factor of safety of approximately 4.

Verification & Validation

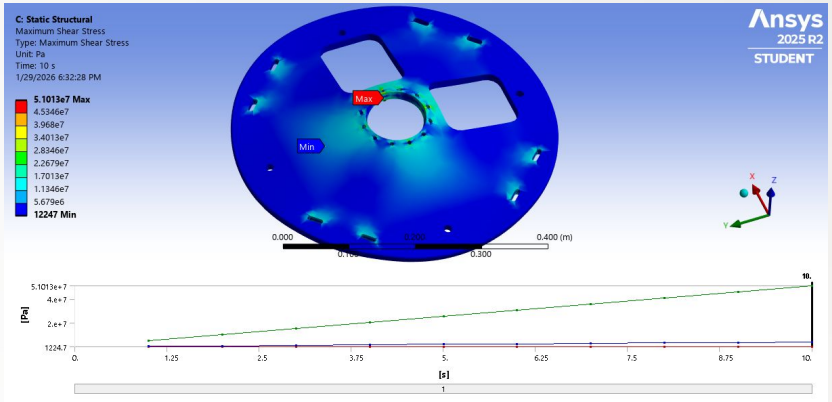
E/D ratio – a critical rule of thumb. If the ratio is too small, it won't fail due to max stress, it will simply fail due to the bolt ripping through the material.

- E is the distance between the center of the bolt hole to the closest edge (this being the giant circle in our base plate), this being 6.55
- D is the bolt hole diameter, this being 6.35 mm

→ 6.55 / 6.35 ≈ 1.03

This is horrible especially when carbon fiber shear strength is 50-70 mPa and the max shear simulation shows 50 mPa

However! Since switching to 6061-T6 Aluminum, this illegal practice can be acceptable due to the shear strength being ≈ 207 mPa. But to ensure safety I reduced the diameter of the center hole by 6 mm for good practice.

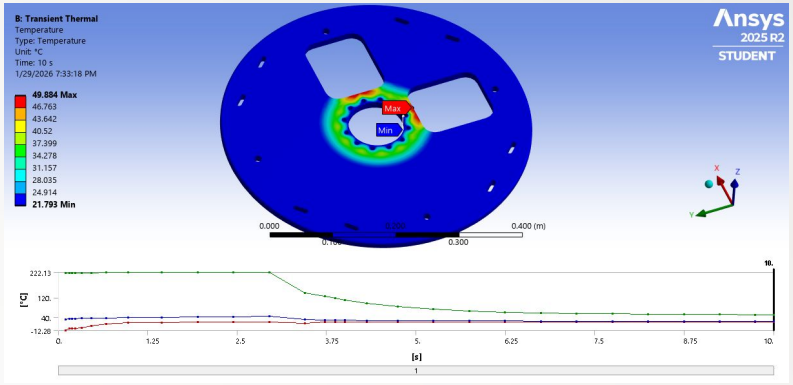


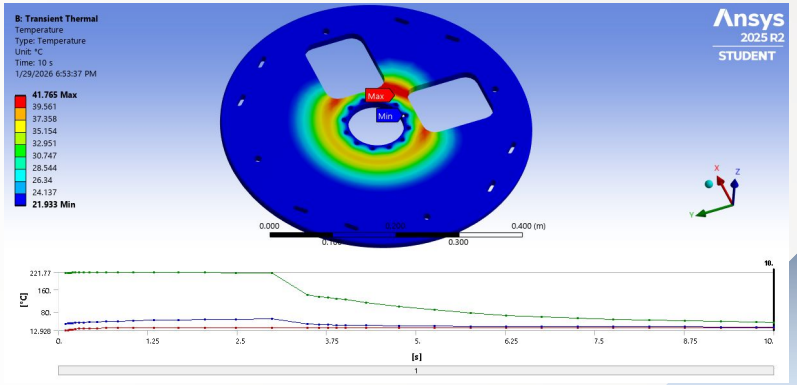
Verification & Validation (Cont.)

Thermally:
With the assuming the heat exerting from propulsion is 220°C (**430°F**)

- Carbon Fiber maximum temperature limit is 82°C (**180°F)** (from its spec sheet) and will lose its structural integrity.
- While Aluminum's limit is 149°C (**300°F**), it does not melt until ~1,100°F, retaining roughly 45-50% of its strength while maintaining a 20% safety factor at max temperature ((35,000 psi)/2) / 15,670 psi (max stress)

Aluminum is also highly conductive dissipating heat across the plate instead of concentrating at the bolt's interface.





Manufacturing & Procurement Plan

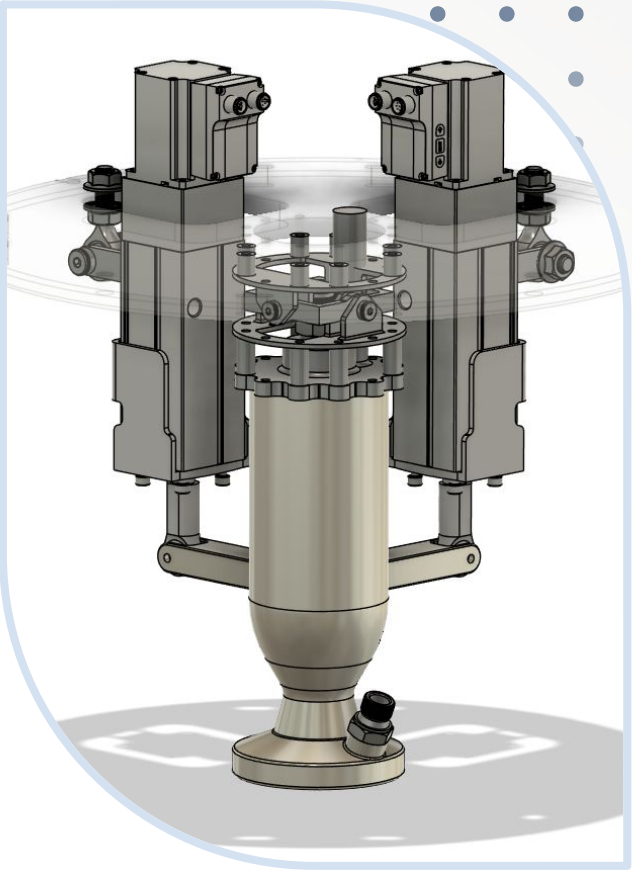
The only components requiring custom machining are the baseplate and the RCS plates. All other structural elements utilize standard stock materials and off-the-shelf hardware, reducing fabrication complexity, cost, and schedule risk.

d	Part	Price	# Quantity	# Pack of	Description	Link
Aluminum Bolt-Together Fran	8809T101	$247.45	4	1	Connecting the bottom plate to the connector plate	McMaster
Aluminum Unthreaded Space	92510A615	$11.61	8	1	Spacers for RCS/Avionics bay area	McMaster
High-Strength Steel Nylon-Ins	90630A150	$4.35	4	20	Locknuts for plate bolts	McMaster
1/4" Aluminum Plates	88895K511	$154.64	3	1	Plates for the frame	McMaster
1-1/2" Long Slotted Corner Br	8681N13	$13.95	32	1	Bracket for the frame	McMaster
Medium-Strength Grade 5 Ste	91247A120	$16.22	2	25	Screws for connecting to square brackets	McMaster
Medium-Strength Grade 5 Ste	92865A109	$16.37	1	50	Screw to hold bracket to plate	McMaster
Medium-Strength Grade 5 S	91247A128	$11.80	1	10	Bolts for RCS spacers	McMaster
Medium-Strength S	93298A125	$7.97	1	50	Flanged lock nuts for bolts	McMaster
6061 Aluminum	9057K43	$624.85	1	1	Aluminum sheet for baseplate	McMaster
Total		$1,109.21				

Interdependencies

To stay on track:

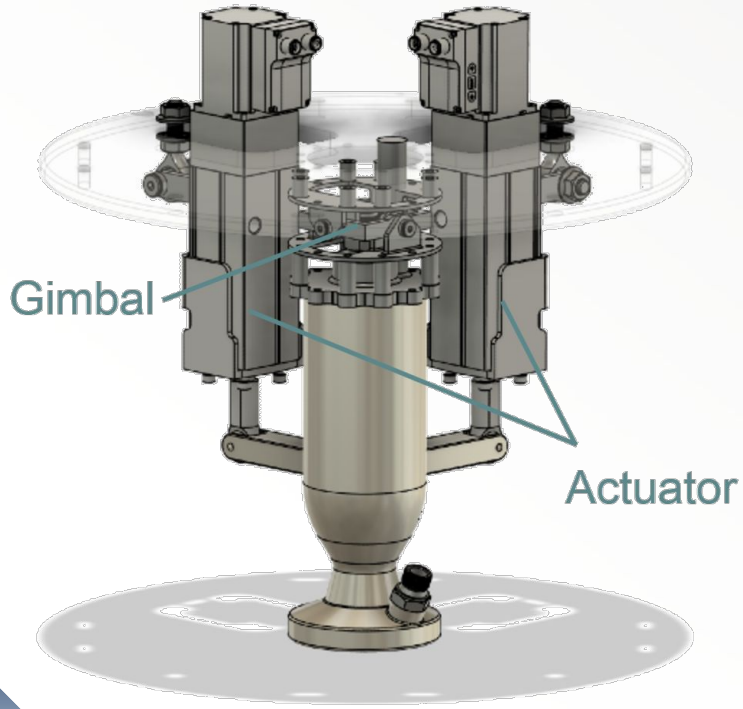
- We need the avionics team to confirm their system dimensions and where they plan to mount it in the vehicle. This will let us know if any structural changes or updates are needed to integrate their system.
- We also need to confirm whether the baseplate and RCS plates can be manufactured in-house or if they will need to be outsourced, as this impacts the final design and timeline.



03

Thrust Vector Control (TVC)

Subsystem Overview



The TVC subsystem is responsible for maintaining rocket verticality during takeoff, flight, and landing by counteracting rotational impulses. In addition, we also need to support the thrust load of the engine.

We do this through a pair of linear actuators in addition to a gimbaled mounting system for the thrust chamber to change the angle of thrust acting on the rocket.

Subsystem Requirement and Specifications

Gimbal Speed

We will aim for the gimbal (and by proxy the thrust chamber) to rotate at a maximum of 30°/sec (0.5 radians/sec)

Gimbal Acceleration

We also have a target acceleration of gimbal at 40 rad/s^2. This can then be extrapolated to a minimum actuator force of ~140lbs.

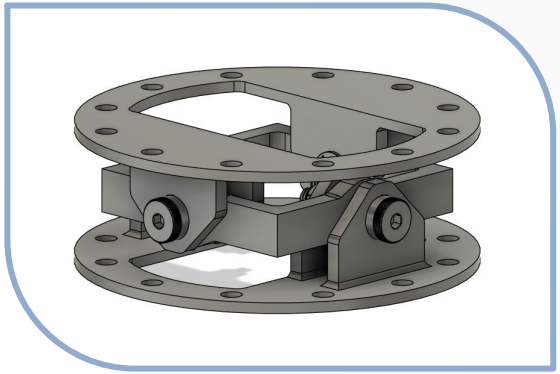
Thrust Load

The nominal load from the thrust chamber is 3kN or roughly 675 lbf.

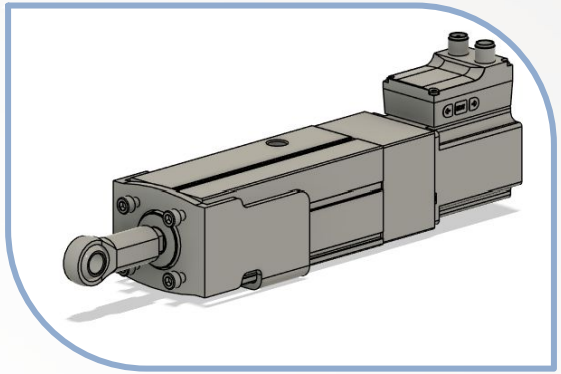
General Optimization

As with all HALO parts, we will look to minimize weight and costs wherever possible. Additionally, we will maximize ease of manufacturing whenever possible.

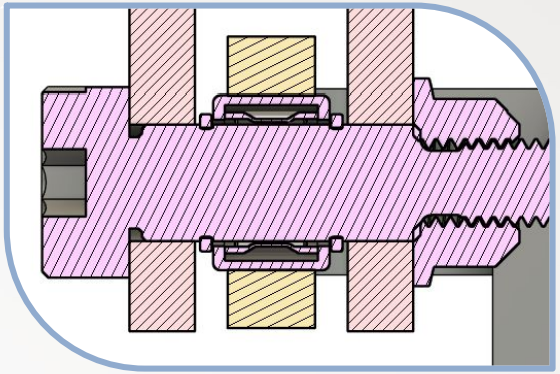
Major Changes Since Last Design Review



Gimbal Redesign



New Actuators

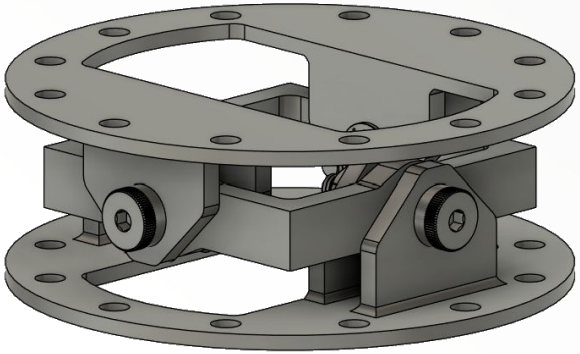


Hardware and Mounting

Gimbal Redesign



New Design



01

-Reduce Footprint Size-

- By shortening level arms, we are able to get a higher loading capacity with less material
- Doesn't require as much stock material, reducing price from material costs and manufacturing

02

-Design Intentionality-

- We add fillets in order to make manufacturing easier
- Less wasted space results in less spacers elsewhere in the design

Gimbal Hardware

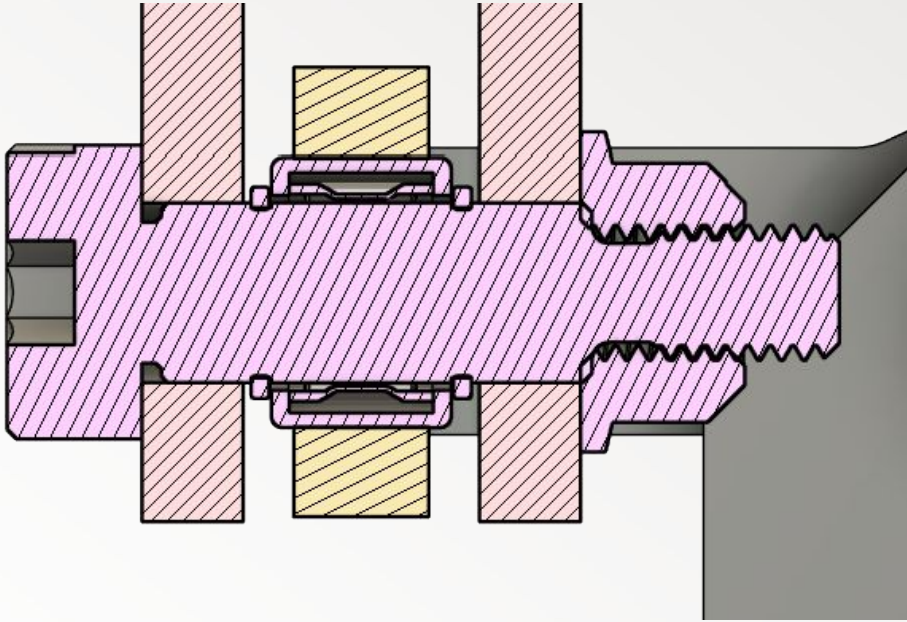
We use a steel shoulder screw to hold the yoke and ring together.

This is paired with a distorted-thread flange locknut for the vibrational resistance

We also use a needle-roller bearing to remove the friction between the ring and yoke. Needle-roller bearings are preferred due to their higher resistance to radial loads.

The inner diameter shaft of the bearing will be held with retaining rings and the outer diameter will be press fit into the ring

Cross Section of the Gimbal Hardware



New Actuators



Re-examining Actuator Specs

In the previous iteration of the TVC we had calculated a minimum actuator strength of 400lbs. This greatly limited our options to only slow, powerful actuators.

Upon re-examining the problem, we agreed this was likely an overestimate of the true value.

We recalculated the numbers to arrive at the new specs:

1. F = ~140 lbf
2. V_{min} = ~2″/sec

Examining Actuators

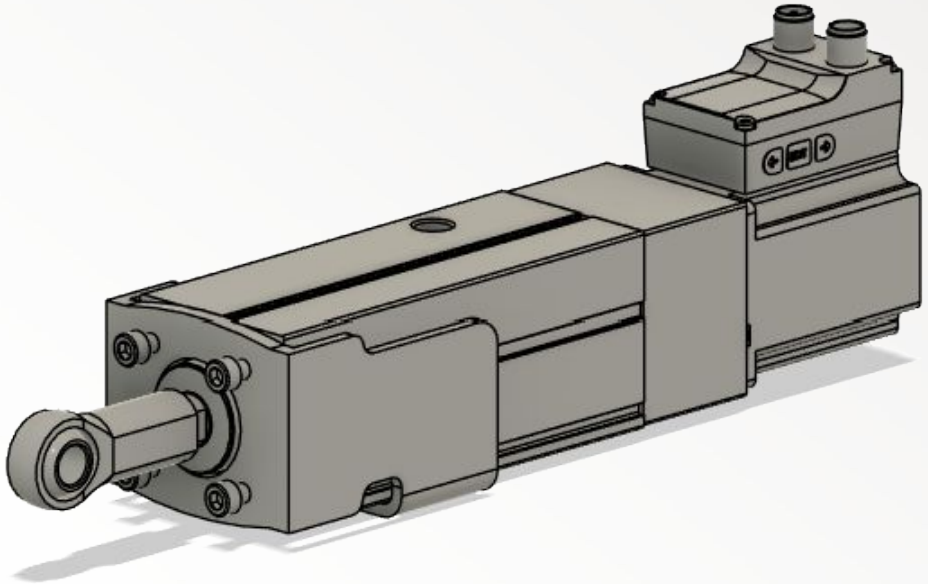
Actuator	Cost	Extension Speed	Force	Total Length	Weight	Stroke Length
Firgelli P-Series High Force Actuators	$154.95	0.3”/sec	400D, 800S	8.58"-12.58"		4”
Firgelli - Heavy duty rod actuators	$149.99	0.49”/sec	400D, 450S	10.37"-13.37"		3”
Progressive Automations - Linear Actuator IP66	$156	2.8”/sec	100D?	9.8”-11.87”	4.6 lb	2”
ServoCity - Super-Duty Linear Accutators	$399.99	2.63”/sec	560D 3000S	15.71”-19.73”	8.89375 lb	4”
Digikey/Festo - EPCS-BS-60-50-5P-A-ST-M-H1-PLK-AA	$1,116.93	3.54”/sec	264.55D 224.81S	12” - 14”	5.817999 lb	2”
Thomson Linear - HD24B017-0150EXX2EESD	$956.23	2.28”/sec	382D 4000S	13.386” - 19”	14.77lb	5.9” (150mm)

Red - Not Feasible/Would require major design change throughout the system

Orange - Not Ideal/Requires sacrifice for weight or price

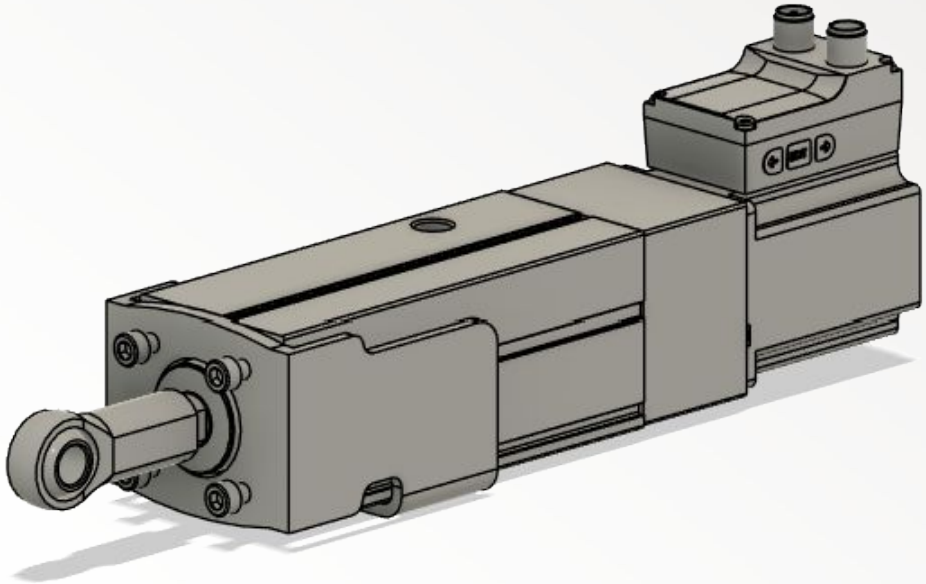
Green - Within Specification

Festo Electric Linear Actuator



- Force : 1000N (~225 lbf) Actuators
- Stroke Length : 50mm (~2″)
- Acceleration : 1.5m/s^2
- Speed : 90 mm/s
- Mass : 2.639 kg
- Price: $1,116.93

Additional Notes



These actuators mount with T-slots, which gives us some more leeway with regards to spacer positioning

Given the high price of this component, we will be looking for a sponsorship in the upcoming weeks

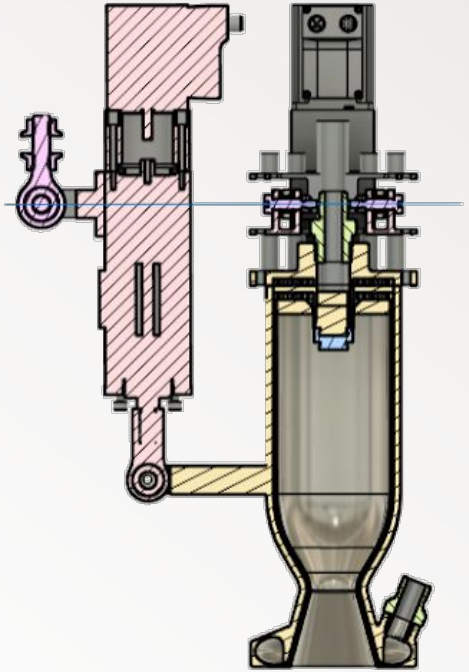
Cross Coupling Problem

The Problem:

- Mounting actuators with optimal speed and force geometries created cross coupling between them
- The coupled system requires more complex control systems and increases the force on the actuators

Our Resolution:

- We aligned our actuator tailstock mount with the COR of the gimbal in order to eliminate the dependencies of the actuators on one another



A Brief Discussion of Materials

We will be using aluminum alloys for all of our custom parts because of its ease of manufacturing and its specific strength

Alloy	Tensile Strength	Weight	Cost
2014 Alloy Aluminum	435 MPa	0.101 lb/in3	$4.10/kg
2024 Alloy Aluminum	469 MPa	0.100 lb/in3	$4.45/kg
6061 Alloy Aluminum	310 MPa	0.0975 lb/in3	$4.00/kg
7050 Alloy Aluminum	552 MPa	0.102 lb/in3	$8.00/kg
7075 Alloy Aluminum	572 MPa	0.102 lb/in³	$4.38/kg

After examining our options, based on our application, we will be using 7050 Alloy Aluminum wherever possible.

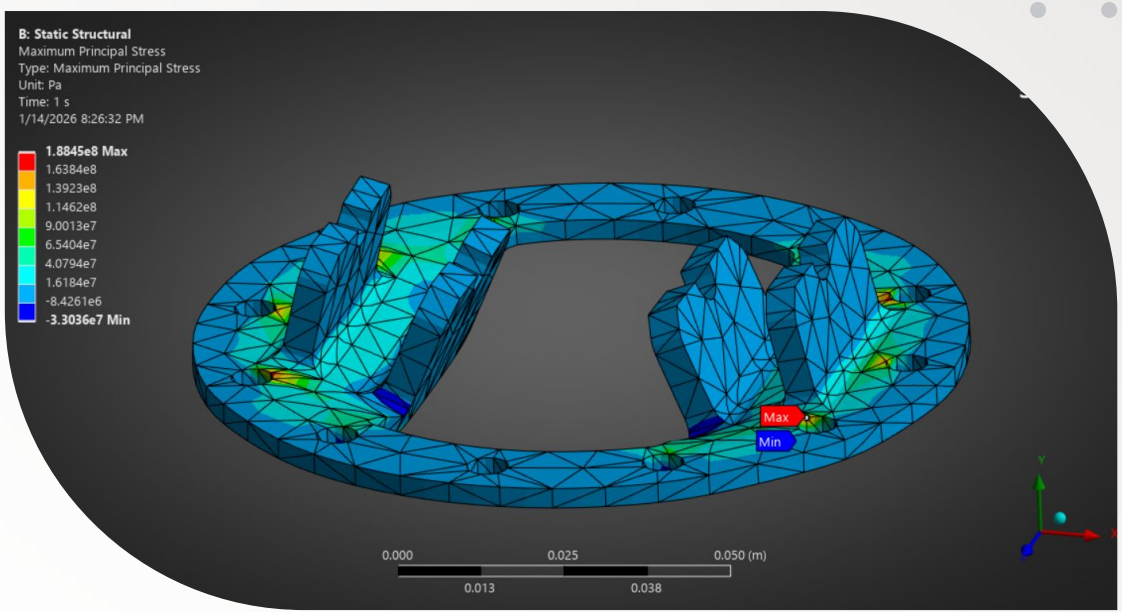
AL 7050 is fairly rare and somewhat expensive. If we are unable to reasonably acquire or afford AL 7050, we will use AL 7075 and AL 2024 as alternatives.

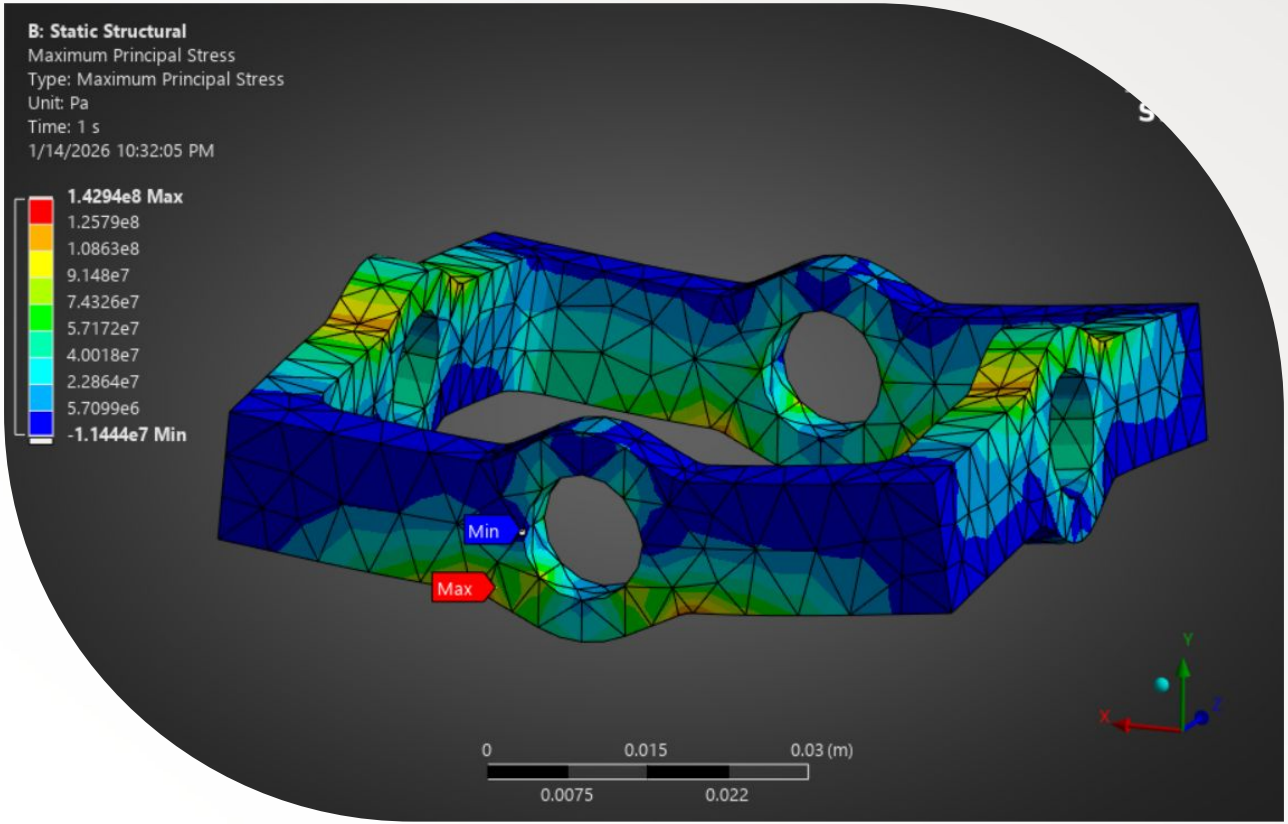
Design Verification – FEA

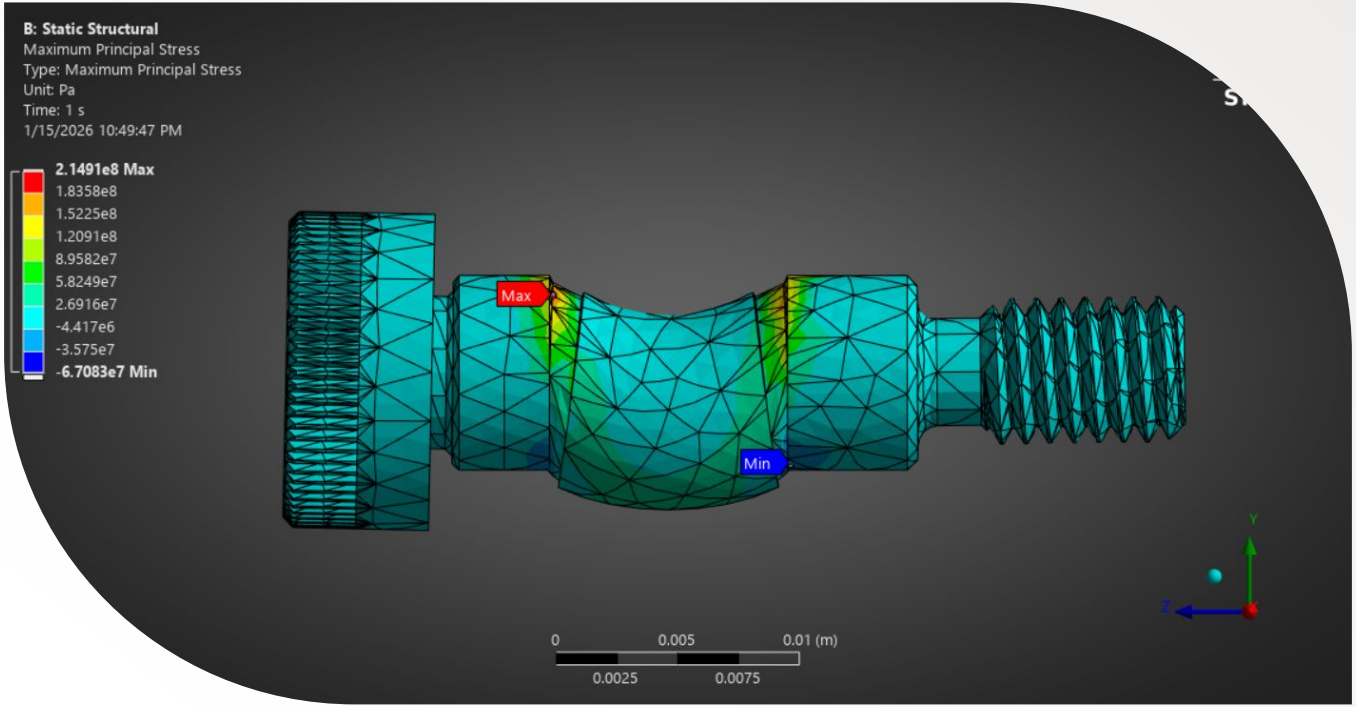
In order to verify our design, we ran each major load-bearing component through Finite Element Analysis in Ansys

All results were calculated using just the loading from the engine. As such we ignored stress that may result from vibrations or thermal flux

We account for these absences with the use of a generous safety factor and conservative estimates




B: Static Structural
Maximum Principal Stress
Type: Maximum Principal Stress
Unit: Pa
Time: 1 s
1/14/2026 10:32:05 PM
1.4294e8 Max
1.2579e8
1.0863e8
9.148e7
7.4326e7
5.7172e7
4.0018e7
2.2864e7
5.7099e6
-1.1444e7 Min
Min
Max
0
0.015
0.03 (m)
0.0075
0.022
Y
X
Z


B: Static Structural
Maximum Principal Stress
Type: Maximum Principal Stress
Unit: Pa
Time: 1 s
1/15/2026 10:49:47 PM
2.1491e8 Max
1.8358e8
1.5225e8
1.2091e8
8.9582e7
5.8249e7
2.6916e7
-4.417e6
-3.575e7
-6.7083e7 Min
Max
Min
0
0.005
0.01 (m)
0.0025
0.0075
Y
Z

FEA Results

Comparing the results for maximum stress against the strength of AL 7050 and AL 7075, we find that both yoke pieces comfortably pass.

As for the screw, the required cuts end up concentrating stress into their locations. While this doesn't outright break the shoulder screw, it is a major informant in our decision to use steel alloy screws as opposed to stainless

Overall, based on the results of this first round of FEA, I'm confident that each part meets the structural requirements that are demanded of them.

BOM

Linear Actuators	EPCS-BS-60-50-5P-A-ST-M-H	$1,116.93	2	1	Linear actuators for TVC, adjust angle of the thruster	Festo
Ball Joints	2458K161	$12.47	2	1	tvc actuators connection to base plate	McMaster
Ball Joint Bolt	91273A469	$11.97	2	1	can buy from home depot not mcmaster	McMaster
8mm M6 Steel Shoulder Scre	92981A203	$2.06	5	1	Shoulder Screws for gimbal, one spare	McMaster
8mm Stainless Retaining Ring	90967A120	$8.24	1	10	Retaining rings to hold bearing	McMaster
M6 Distorted Thread Locknut	3528N21	$6.49	1	25	Locknuts, distorted thread can only be applied once	McMaster
8mm Needle Roller Bearing	5905K73	$4.88	5	1	Needle Roller Bearings for gimbal. Once Spare	McMaster
8mm Shoulder Shortening Sp	92927A912	$9.29	1	10	In case we need to move shoulders out of the way of propulsion	McMaster
Gimbal Stock Material		$394.00	1	1	Stock Material of AL 7075. Does NOT include manufacturing cost	
Total		$2,735.46				

*A few fasteners are missing from this list

Future Plans and Roadmap

01 Affording Actuators

- As mentioned before, the new actuators are uncomfortably expensive
- We will be looking to acquire a sponsorship from The Festo Corporation in order to ease this cost

02 Buy Parts

- As soon as possible we will begin to order parts so that we may begin further testing

03 Better FEA

- As parts are arriving, we will reevaluate the FEA
- In more sophisticated model we will consider vibrations, thermal stress, and harmonics

04 Assembly and Verification

- Once all parts have arrived, we will begin assembling and testing the subsystem



04

Landing Legs

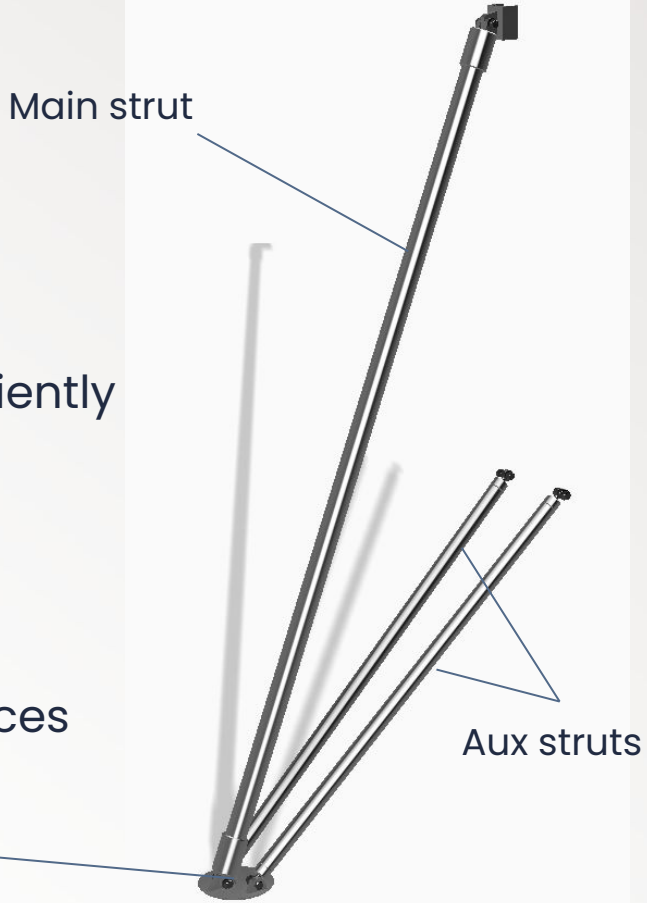
Subsystem Overview

Mission Success Criteria:

- Support weight of 200 kg vehicle before takeoff
- Absorb landing loads during vertical touchdown
- Transfer impact load into primary structure efficiently
- Minimize system mass

Architecture:

- Multi-strut, splayed landing leg system
- Al 6061 Tubing and custom end fixtures
- Pinned joints at rocket body and footpad interfaces



Technical Specifications

System Geometry

- Main Struts: **2.1 m (7 ft)**
- Auxiliary Struts: **1.2 m (4 ft)**
- Splay angles: Main **22°** Auxiliary **42°**
- Landing Footprint: **25 ft^2**

Tubing Sizes

- Main Struts: **2” OD, 1.5” ID**
- Auxiliary Struts: **1.5” OD, 1.25” ID**
- Thickness: **0.25”**

Material and Mass

- Struts + fixtures: **6061 Aluminum**
- Pins / fasteners: **High strength steel**
- Mass per 3-leg subsystem ~ **7 kg**
- Total leg mass ~ **28 kg**

Fixtures and Interfaces

- Welded aluminum end fixtures
- Plug + clevis style interfaces
- Bolted shear connections

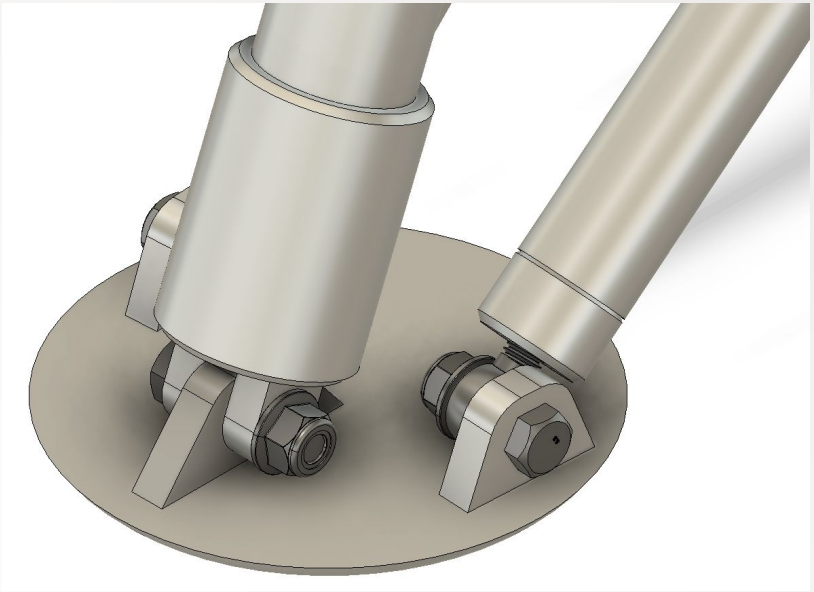
Strut Interfaces

Base Plate Connector



- Clevis-style pinned joint to vehicle structure
- Compliant with TVC / engine architecture

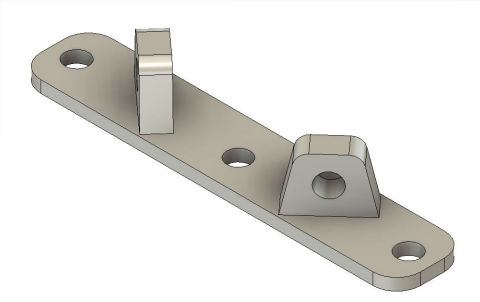
Foot Pad Connector



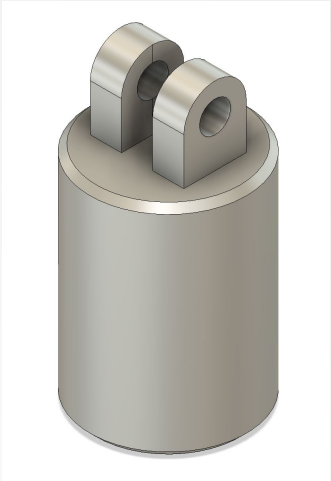
- Pinned lug–fork interface with shear-loaded bolt
- Permits rotation while reacting to landing loads

Custom Parts

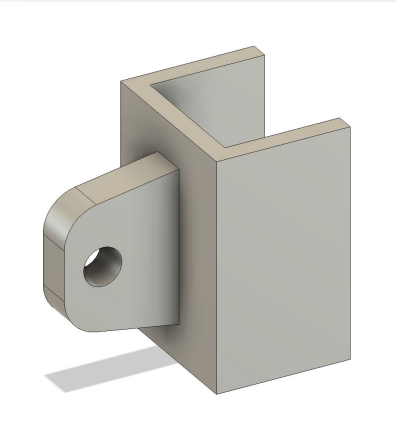
Auxiliary Strut Connector



Main Clevis Cap



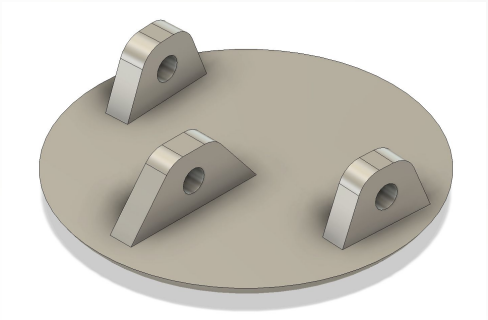
Main Strut Bracket



Auxiliary Weld Bolts



Foot Pad



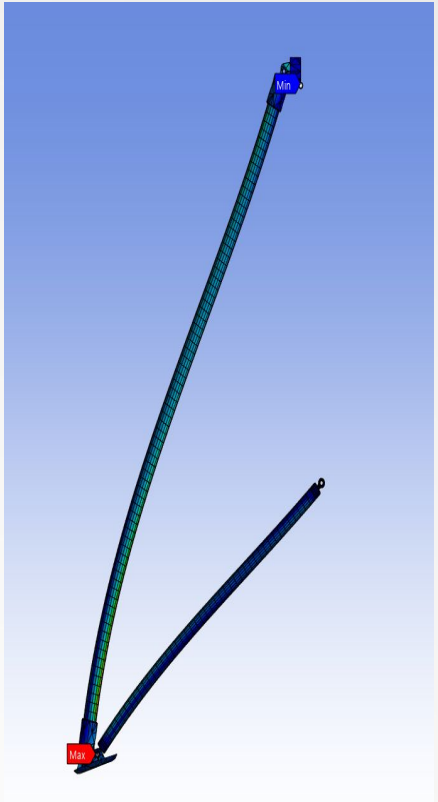
Validation - FEA Setup

Model:

- Static structural analysis because of computation constraints (ANSYS)
- Single 3-leg subsystem analyzed for worst case scenario

Constraints and Loading:

- Fixed constraints at all 3 connection points to rocket structure
- 2000N vertical force applied to footpad base



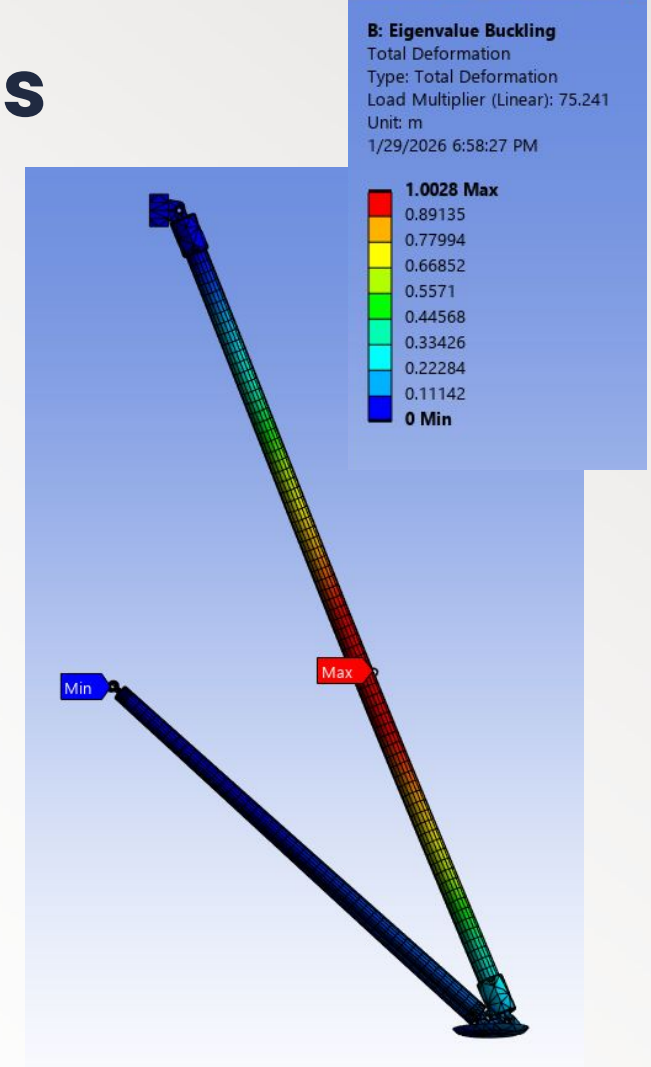
FEA Results – Buckling Analysis

Results:

- Linear buckling analysis predicts a first mode buckling at a load multiplier of **75x**

Interpretation + Takeaways:

- Buckling occurs well above static load
- Given extremely high FoS, buckling likely won't be failure mode
- Potential to reduce tubing thickness (leading to reduced mass)



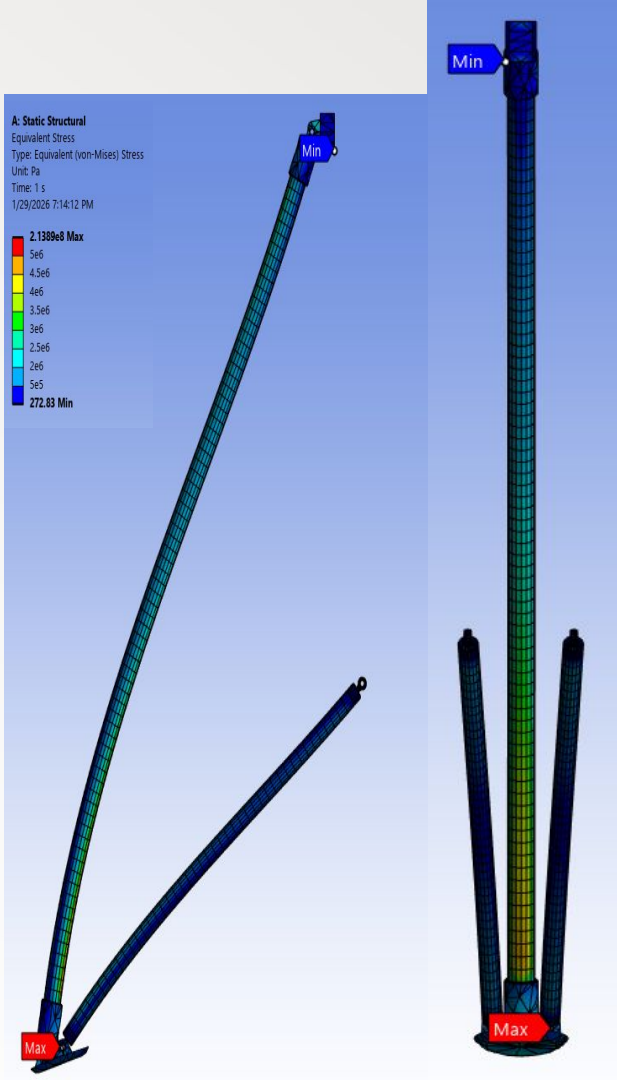
FEA Results – Stress Analysis

Results:

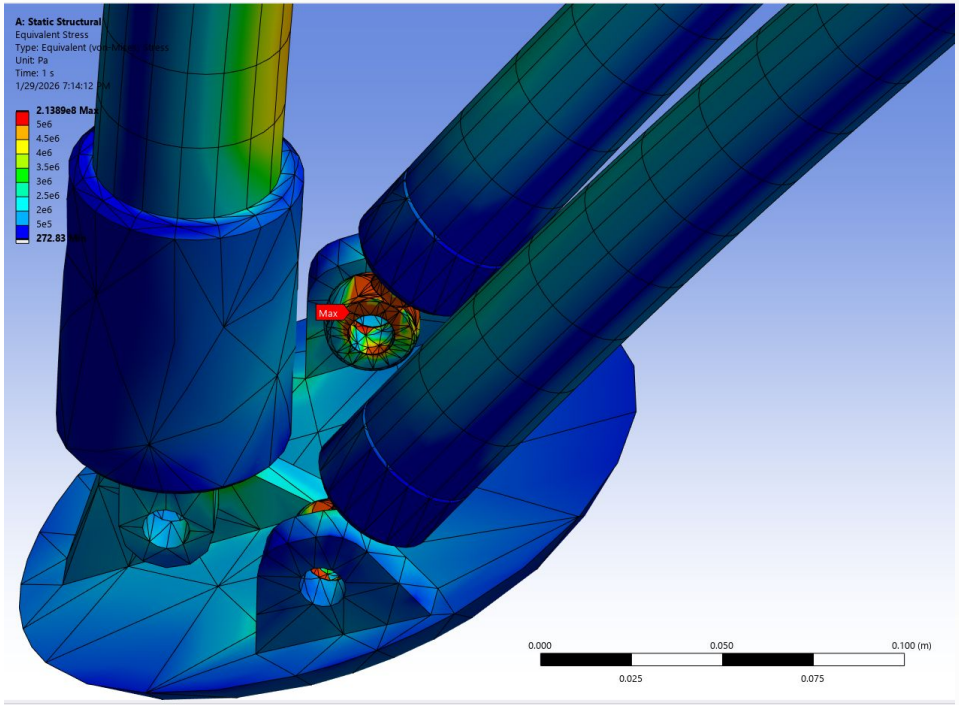
- Main struts experience low axial and bending stresses
- Aux struts also well below yield strength

Interpretation + Takeaways:

- Load transfer dominated by axial compression (success!)
- Stress concentrations at interfaces will likely govern failure



FEA Results – Stress Analysis cont



Issue:

- Extreme stress concentration at footpad fixtures ~ FoS 4

Design Response:

- Iterate on fixture geometry to eliminate stress risers
  - Increase contact area
  - Fillet transitions
  - Switch to double shear configuration
  - Add collar or reinforcing sleeve

BOM

Part	Quantity	From	Total Price	Manufacturing Cost
1.5" 1/2-13 Extreme Strength Hex Bolt 90201A418	12	McMaster	$30	None
2" 1/2-13 High Strength Hex Bolt 90002A125	8	McMaster	$21	None
High Strength Washer 92018A113	10	McMaster	$50	None
1/2-13 High Strength Flange Nut 92018A113	24	McMaster	$30	None
2440K82 High Strength Threaded Rod End Bolt	8	McMaster	$200	None
6061 2" OD 1.5" ID	28 ft	Online Metals	$1,000	None
6061 1.5" OD 1.25" ID	36 ft	McMaster	$720	None
6061 2.5" OD 2" ID	4ft	Amazon	$120	None
Stock 6061 Aluminum for Manufacturing	4ft^2+	McMaster	~$500	Need Quotes

~ $2701

Interdependencies + Looking Forward

Immediate Next Steps:

- Iterate and validate footpad fixture geometry → look to optimize mass by potentially reducing tubing thickness
- Effective forces static modeling using avionics data
- Higher fidelity modeling - dynamic simulations

Other subsystem interdependencies:

- Small tweak to TVC mounting geometry
- Verify the structure can handle impact reaction loads
- Propulsion engine clearance spec?

Questions?